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Muscle activation and length changes during two lunge exercises: implications for rehabilitation

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Eccentric exercises are commonly used as a treatment for various muscle and tendon injuries. During complex motions such as the forward lunge, however, it is not always clear which muscles may be contracting eccentrically and at what time. Because this exercise is used during rehabilitation, the purpose of this investigation was to determine what type of contractions take place during two different types of forward lunge and assess the implications for rehabilitation. Five experienced athletes performed five cycles for each of the walking and jumping forward lunges. Motion analysis was used to calculate the shortening or elongation of each muscle based on the change of position of their origin and insertion points during the lunge. Electromyography of the lateral hamstrings, rectus femoris and lateral gastrocnemius was combined with the muscle length change data to determine when isometric, concentric and eccentric activations occur during the lunge. Eccentric contractions in both the quadriceps and gastrocnemius were observed during the lunge. No hamstring eccentric contractions were found; however, the hamstrings showed isometric contractions during the first part of the stance phase.

Over the past few years, different kinds of eccentric exercises have been developed and used for prevention and rehabilitation of musculo-tendinous injuries. For example, eccentric calf exercises are now commonly used for treating Achilles tendinopathies (Alfredson et al., 1998; Silbernagel et al., 2001; Roos et al., 2004; Shalabi et al., 2004). Eccentric training exercises with a specially designed training device are also used for prevention of hamstring injuries (Askling et al., 2003). Given the success of these methods, it is desirable to find more exercises that could be used for the prevention and treatment of musculotendinous injuries and, in particular, exercises that do not require specialized equipment. Based on anecdotal evidence from athletes and trainers, these exercises may already be in practice. However, the effectiveness of some of these exercises, and what these exercises may achieve from a biomechanical perspective, is not always clear.

One such exercise is the forward lunge, which is widely used in the athletic community during training and rehabilitation and requires no specialized equipment. Different kinds of forward lunges have long been used by coaches and their athletes, particularly in the track and field community. These exercises are seen as very strenuous and are thought to strengthen the thighs, especially the hamstring group, eccentrically. Forward lunge is also a closed kinetic chain exercise (Pincivero et al., 2000), in contrast to exercises commonly performed in isokinetic dynamometers such as leg flexion-extensions. In addition, forward lunges have recently been used to identify differences in the movement pattern of two different types of anterior cruciate ligament (ACL) patients (Alkjaer et al., 2002; Mattacola et al., 2004). It would therefore be beneficial to understand how variations of the lunge affect muscle activations and thus lead to its more appropriate use as a training and rehabilitation intervention.

Even though eccentric exercises have been widely discussed in the scientific community, its definition is somewhat confusing (Faulkner, 2003), with definitions that may vary depending on the movements being observed (Jonhagen et al., 1994; Eston et al., 1994; Byrne et al., 2001). For the purposes of this investigation, eccentric contraction is defined as an elongation of the observed muscle while this same muscle is active. Given the complexity of the lunge movement from a functional anatomy perspective, it is difficult to determine whether the hamstring

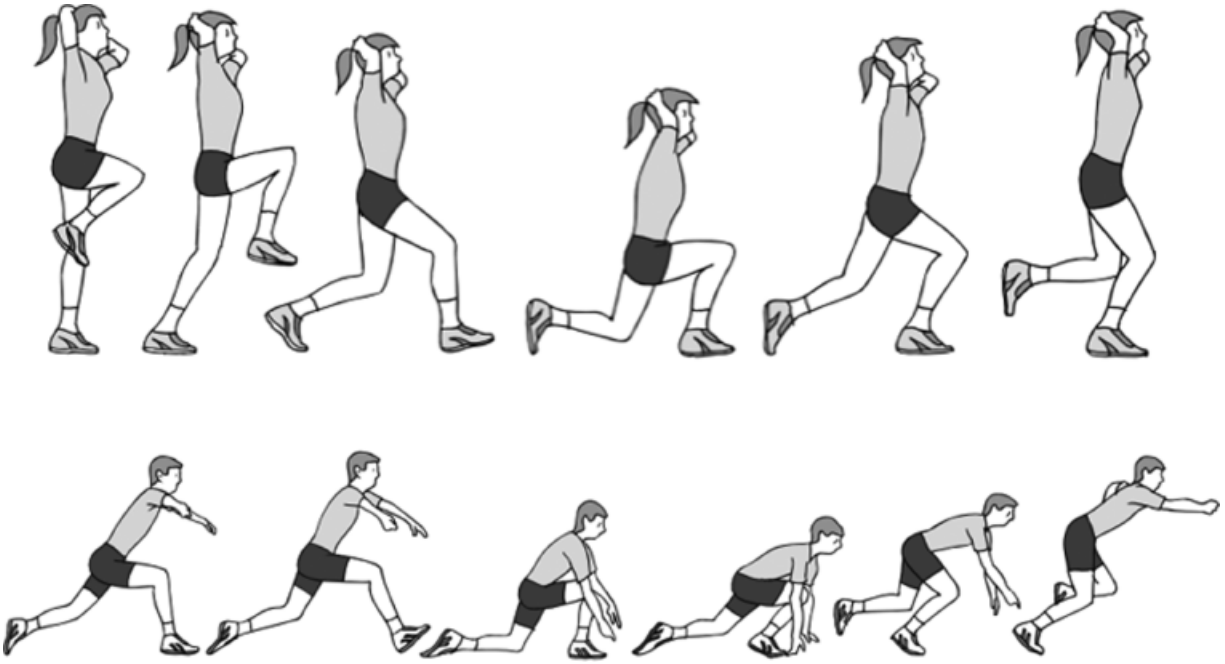


Fig. 1. Walking (top row) and jumping (bottom row) forward lunge exercises. Note the different positions of the upper body during the exercises. The JFL starts in a sprint start position with the right foot in front. During the JFL the hip and upper body is leaning more forward than in the WFL.

muscles are acting eccentrically during the lunge from observation alone. However, previous investigators have used motion analysis to determine changes in muscle length during knee flexion-extension (Hawkins & Hull, 1990). When this is combined with electromyography (EMG) to determine muscle activations, it is possible to identify when muscles may be contracting concentrically or eccentrically during movement.

The purpose of this investigation was to perform a descriptive analyses of two types of forward lunge: the walking forward lunge (WFL) and the jumping forward lunge (JFL) (Fig. 1), in order to gain an insight into the role of the leg muscles in performing these movements. It was hypothesized that during both the WFL and JFLs, the quadriceps, hamstrings and gastrocnemius muscles would act eccentrically; however, the jumping lunge would elicit greater eccentric activation of the hamstrings than the walking lunge.

Material and methods

Subjects

Five experienced male athletes with a mean age of 31 (range 27–39) years, all skilled in different types of forward lunges, were asked to participate in this study. All subjects gave their informed consent and the Ethical committee at the Karolinska Institutet approved the study (03-776).

Study protocol

The subjects performed warm-up exercises for 15 min that included jogging and mild stretching of the quadriceps, hamstrings and gastrocnemius. Subjects were then informed of the appropriate techniques and asked to perform a few forward lunges on the track to familiarize themselves with the correct motion. Subjects were then brought into the lab and prepared for the EMG and motion analysis recordings. The movement protocol consisted of the subject performing one full cycle of a WFL five times. During each cycle, the subject was asked to step onto the force platform with their left leg. The same procedure was repeated for the JFL. The subjects started the WFL standing with their feet parallel and thereafter began stepping forward with the right leg. The next step was onto the force platform with the left foot making contact. The subject started the JFL as a sprint start with the right foot in the front and again stepping onto the force plate with the left foot (Fig. 1).

Table 1. Positioning of surface electrodes

Muscle:	Location of electrodes
Lateral hamstrings	50% on the line between the ischial tuberosity and the lateral epicondyle of the tibia
Rectus femoris	50% on the line from the anterior spina iliaca superior to the superior part of the patella
Lateral gastrocnemius	At the proximal 1/3 of the line between the head of fibula and the heel

Muscle activity was measured using EMG (Myosystem 2000®, Noraxon Inc., Scottsdale, Arizona, USA) from three muscle groups (lateral hamstrings, rectus femoris and lateral gastrocnemius). EMG was recorded at 1000 Hz with a frequency bandwidth of 16–500 Hz and a common mode rejection ratio of 114 dB. The skin was shaved and cleaned with alcohol. Pregelled bi-polar Ag–AgCl surface electrodes were placed with an inter-electrode distance of 20 mm and placed parallel to the muscle fibers on the muscle bulk. Electrode locations are described in Table 1 and can be seen in Fig. 2.

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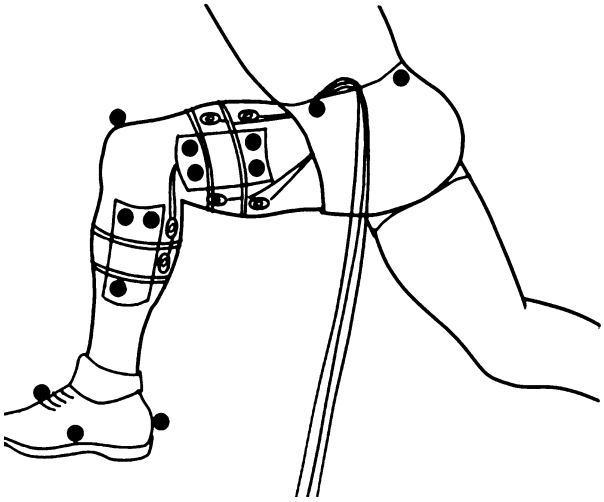


Fig. 2. Experimental set-up. EMG-electrodes and cables. Spherical reflectivemarkers.

Table 2. Electromyography signals were recorded during maximal voluntary isometric contraction for each muscle using the following joint positions

Muscle	Exercise
Lateral hamstrings	Subject prone on the floor. The hip at 0°, and the knee flexed 60°. The lower leg is externally rotated. Subject attempts flexion of the lower leg
Rectus femoris	Subject sits on a bench with the thigh fixed and the knee flexed 60°. Subject attempts extension of the knee
Lateral gastrocnemius	Subjects stands under a fixed bar with the knee flexed 20° and attempts extension of the ankle joint

The EMG signals were AD converted and stored on a computer for future analysis. The signals were root mean square converted with a time constant of 50 ms. Maximal voluntary isometric contractions (MVIC) were performed for each muscle group (Table 2) and the EMG of the lunge cycle was expressed as a percentage of the MVIC value for each muscle (% max EMG).

Motion analysis

Clusters of lightweight ∅ 19 mm reflective markers were placed on the pelvis, the left thigh, the left lower leg and the left foot (Fig. 2). Four markers were attached on the pelvis with double-sided tape on the anterior and posterior iliac spines. On the thigh and lower leg, four and three markers, respectively, were attached to lightweight, padded aluminum plates, which were taped to the segment. A single marker was placed on the patella. At the foot, markers were anchored to the heel, the midfoot and the distal part of the first metatarsal on the lateral side. The marker trajectories were recorded using an eight-camera motion capture system (ProReflex, Qualisys Medical AB, Gothenburg, Sweden) sampling at 100 Hz. The cameras were placed in a semi-ellipse with the force platform in the center. The calibrated measurement volume was approximately 5 m long, 3 m wide and 2 m high.

Table 3. Each muscle'ss origin and insertion were defined in relation to a fixed body segment

Muscle	Origin	Insertion
Lateral hamstrings	Pelvis (tuber ischi)	Lower leg (caput fibulae)
Rectus femoris	Pelvis (spina iliaca ant superior)	Lower leg (the upper pole of patella)
Lateral gastrocnemius	Thigh (post lat fem condyle)	Heel (tubercle of calcaneus)

The 3D movement of the body segments was then computed from the marker trajectories using the methods of Soderkvist and Wedin (1993). The exercises were also filmed by a digital video camera, which was placed parallel to the exercise track.

The origins of lateral hamstrings, rectus femoris and lateral gastrocnemius muscles were identified by a custom-made optical pointing device. The device has two reflective markers with an adjustable distance in between. The device is used to indicate a point in space that lies on the line going through the two markers at a position that is defined by the distance between the two markers (Cappozzo et al., 1995) Using the device, the estimated locations of the origin and insertion of the observed muscles were indicated. An experienced orthopedic surgeon palpated and marked the anatomical positions used to determine the muscle origin and insertion points (Table 3).

Analysis of captured motion data was performed with custom-written Matlab (The Mathworks, Natick, Massachusetts, USA) code. The straight-line distance between the origin and insertion points of each muscle approximated the muscle lengths and determined muscle length changes throughout the motion cycle. This was used to determine whether the muscles were lengthening or shortening during the motion cycle. The mean values of muscle length changes were calculated for each person and ensemble averaged.

Force platform

Ground contact was recorded with a force platform (AMTI Inc., Watertown, Massachusetts, USA). The analog data signal corresponding to the vertical component of the force data was split and fed into both the EMG and motion capture systems for synchronization. The signal was also used to determine the duration of the stance phase. The force plate data were sampled at 1000 Hz.

Data presentation

All data were time normalized to 100% of the lunge cycle from foot strike to toe-off using 100 data points. An additional 10% (and 10 data points) was considered before foot strike, which we will call the ''approach'' phase of the lunge movement and the mean muscle activation values and muscle length changes were calculated for each cycle. For JFL, the stance phase is the interval from foot strike to toe-off of the left foot. In the case of WFL, the start of the stance phase is also defined as foot strike; however, the end of the stance phase is defined differently. The left leg remains in contact with the force platform for a period after the foot strike of the right foot, and the force under the left foot exhibits a characteristic drop as the center of mass moves toward the right foot. We define this abrupt decrease as the end of the stance phase for the WFL.

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Results

All five athletes completed the experiments without any complications and did not report any difficulties in performing the lunge tasks. A number of differences were observed between the jumping and WFL. While the muscles of the RF, LH and LG were active during all phases of both lunges, the JFL consistently elicited a higher activation than the WFL (Fig. 3(a)–(c)). As expected, a greater range of motion at the hip, knee and ankle joints was found during the JFL as compared with the WFL (Fig. 3(d)–(f)). The altered kinematics between the JFL and WFL resulted in longer RF and LH muscle lengths while the LG lengths appeared to be more consistent across the two motions (Fig. 3(g)–(i)).

Rectus femoris

During the beginning of the stance phase, there is a slight elongation of the rectus femoris (Fig. 3(g)) preceded by a high activity in EMG (Fig. 3(a)). For both the WFL and JFL, there is a burst of EMG immediately following foot strike. In the WFL, this is followed by a second burst beginning at 20% stance and peaking at approximately 40% of the stance phase. This coincides with a slight shortening of the muscle, followed by a lengthening of the muscle for the remainder of the stance phase.

In the JFL, the second burst begins at 20% stance and is prolonged throughout the remainder of the stance phase, with the relatively high activity peaking at approximately 60% stance (Fig. 3(a)).

During the first part of the stance in the jumping lunge, there is a distinct increase in muscle length, preceded by a peak in muscle activity, thus representing an eccentric contraction. This muscle lengthening is then followed by a gradual concentric contraction, followed by an eccentric contraction at the end of the stance phase. Note that this transition from concentric to eccentric contraction occurs with relatively high and constant EMG activity just before push-off.

During the WFL and JFL, the rectus femoris muscle was contracting eccentrically for $54 \pm 9\%$ and $44 \pm 8\%$ of the lunge cycle, respectively, from the approach phase to toe-off.

Lateral hamstrings

The EMG data indicate two peaks of muscle activation: one just after foot strike and one just before push-off (Fig. 3(b)). During the JFL, the length of the muscle is almost constant at the first peak, which indicates an isometric contraction, followed by a marked shortening of the muscle until the end of the lunge cycle (Fig. 3(h)). The pronounced second EMG peak for the jumping lunge coincides with the hip extension of the push-off phase (Fig. 3(e)).

During the WFL, there is a short period at the beginning of the stance phase with elongation of the hamstring muscle (Fig. 3(h)) preceded by a high EMG activity (Fig. 3(b)) that indicates a short period of eccentric hamstring contraction.

During the WFL and JFLs, the hamstring muscles were contracting eccentrically for only $18 \pm 8\%$ and $19 \pm 8\%$ of the lunge cycle, respectively, from the approach phase to toe-off.

Lateral gastrocnemius

For both the JFL and WFL, there is an initial small burst of EMG activity at foot strike (Fig. 3(c)) that coincides with a rapid decrease in gastrocnemius muscle length (Fig. 3(i)). During the JFL, this burst is followed by a progressive increase in EMG peaking at approximately 45% stance, while the muscle length gradually increases throughout the stance phase and until toe-off.

During the WFL, there is also a gradual increase in EMG peaking slightly later at 50% stance. Note that the EMG activity is significantly lower than during the JFL and decreases immediately following this mid-stance burst before rising again at toe-off. For the WFL, there is a period of constant muscle length at 80–100% of the stance phase and relatively low levels of EMG; this corresponds to the transition phase from one leg to the next.

During both the WFL and JFL, the lateral gastrocnemius muscle was contracting eccentrically for $61 \pm 3\%$ and $63 \pm 12\%$ of the cycle, respectively, from the approach phase to toe-off.

Discussion

The purpose of this investigation was to gain an insight into the role of the leg muscles in performing forward lunge movements. We investigated two types of forward lunge, the JFL and WFL, and found that although these motions are generally believed to elicit eccentric activations of the hamstring muscle groups, according to our methodology this was only the case during 18% and 19% of the lunge cycles, respectively. However, both the JFL and WFL did elicit longer periods of eccentric contractions for both the RF (44% and 54% of the lunge cycle, respectively) and LG (63% and 61% of the lunge cycle, respectively) muscle groups. We estimated muscle length changes by combining rigid-body kinematic analysis using bone-embedded frames of reference (Cappozzo et al., 1995) with a custom-made optical pointer device used to define the origins

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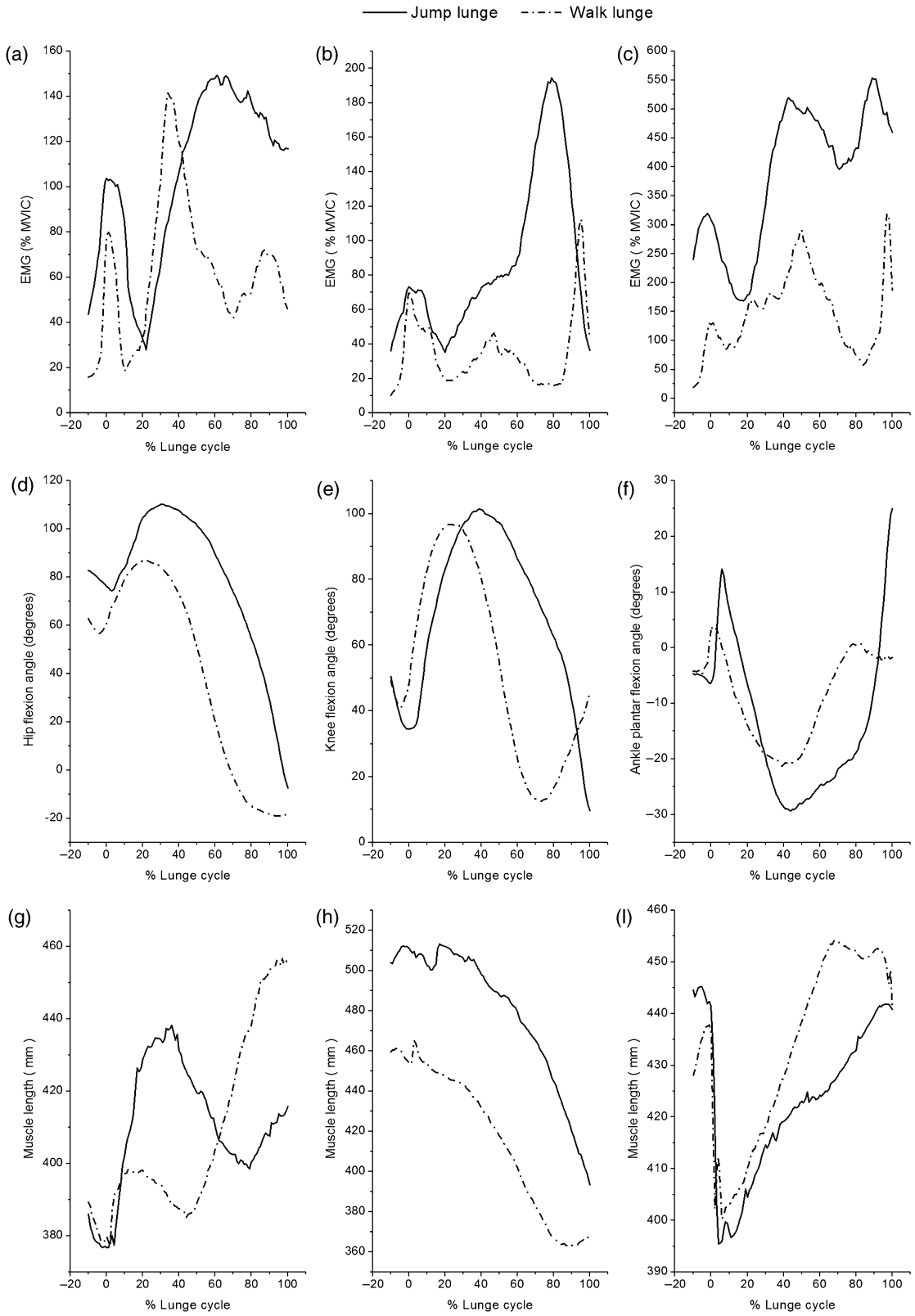


Fig. 3. EMG as percent of maximal voluntary isometric contraction (%, dotted line) and muscle length (mm, filled line) during one cycle of jumping forward lunge (A above) and walking forward lunge (B bottom). The rectus femoris (RF), lateral hamstrings (LH) and lateral gastrocnemius (LG) are displayed from left to right. Heel-strike occurs at 0% and toe off at 100%.

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of each muscle in relation to the bone frames. The muscle origin and insertion were assumed to be points and the lengths of the muscles were approximated as the linear distance between these points. In reality, the paths of the muscles are curvilinear, and the contraction along the path of the muscle may not be uniform. Nevertheless, for the purposes of this investigation, we believe this to be a useful approximation of muscle lengths (Hawkins & Hull, 1990).

During the first part of a forward lunge, negative work is performed as the downward vertical momentum has to be decreased (Pincivero et al., 2000). We found that the hamstring muscles had a very short and rapid period of elongation during the first part of the WFL. In the first part of the JFL, the ensemble-averaged hamstring length remained relatively constant, indicating a quasi-isometric contraction. Individual subject data indicated eccentric hamstring contraction during the WFL or JFLs accounting for 18% and 19% of the total lunge cycle from approach to toe-off. Being two-joint muscles, the hamstring length depends on both knee and hip joint angles. The angle of the hip has more impact on the length of the hamstrings than the angle of the knee (Hawkins & Hull, 1990), due to the longer moment arm at the hip, and this relation increases with increasing knee angle. Small changes in the hip angle may compensate for changes in knee joint angle, which may explain why there is an isometric contraction of the hamstrings during the early lunge phase even though the knee flexes at this time.

In the rectus femoris, we found two periods during the WFL when the muscle elongates while active: at foot strike and at the end of the movement. The contraction characteristics are somewhat different in the jumping lunge: in the beginning of the foot strike, there is a marked elongation of the rectus femoris, followed by a sustained period of concentric contraction. During the first part of this elongation, the EMG activity is high, possibly due to the need to absorb the higher impact forces sustained during heel-strike for the jumping lunge.

The gastrocnemius muscle seems to have a similar pattern for both lunge motions, as evidenced by the similar amount of eccentric contraction (61% and 63%) and as shown in Fig. 3. At the very beginning of stance, there is an elongation of the muscle with a low level of muscle activity. There is then a prolonged elongation throughout the stance phase. The EMG shows two peaks: one in the middle of the stance phase and one at the end. The peak EMG in the middle of the stance phase may represent an eccentric contraction in which the contractile components stretch the series elastic components in preparation for the concentric contraction in the push-off that follows; however, this cannot be verified using the current methodologies. It must be noted that the gastrocnemius activation levels observed in this study were generally high. This could be explained by the method we used to measure the maximal voluntary isometric contraction. We believe that the method in our study allows only relatively low levels of activation to be reached and thus our method may not be optimal to achieve maximal activation of the gastrocnemius muscles. That said, given the fact that both lunges used the same normalization method any difference in activation caused by the movement patterns would be detected.

Ito and co-workers have studied the muscle mechanics in the tibialis anterior muscle complex (Ito et al., 1998) during an isometric contraction. They found by ultrasonography that there is a non-isometric behavior of the muscle fascicles. During isometric contraction, the muscle fibers shorten while the tendinous complex elongates. Hof described a similar phenomenon after studies of the triceps surae muscle tendon complex during running (Hof et al., 2002). He also found that during an eccentric contraction where the musculotendinous complex lengthens, the muscle (contractile component) shortens and the tendon (series elastic component) elongates. EMG activity is normally followed by an electromechanical delay before force generation (Simonsen et al., 1985); however, this delay would only occur in a muscle that is not under tension (fully inactive). In this study, it can be observed that all three muscle groups are active to some degree throughout the lunge cycle. It is therefore likely that any change in muscle length and/or activity would have an immediate effect on the forces generated by that muscle. We have observed a quasi-isometric contraction with a relatively high muscle activity of the hamstrings during the initial phases (approximately 0–20%) of the forward lunge. During this phase where only small changes in muscle length are observed, it is possible that contraction causes shortening of the contractile component while lengthening the parallel and series elastic components with little observable change to the muscle-tendon unit length.

The cyclical motion of the both forward lunges will cause stretch-shortening cycle contraction, as evidenced by the changes in the direction of the muscle-length curves in Fig. 3. This stretch-shortening action has been associated previously with a storage of elastic energy (Jacobs et al., 1996) and a significant increase in muscle work output (Benoit and Dowling, 2005). While it is clear from our investigation that the hamstring, rectus femoris and gastrocnemius muscles will undergo a stretch-shortening cycle during both the WFL and JFL, examining how the performance enhancement of the stretch-shortening cycle may contribute during the forward lunge is beyond the scope of this investigation.

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In this study, EMG was normalized to maximum voluntary isometric contractions, a common method that has also been used to analyze the running sprint (Jonhagen et al., 1996). This method has been shown to be more sensitive in detecting changes in activation in ACL-injured subjects (Benoit et al., 2003) and therefore seems appropriate for this investigation because the lunge may be a useful tool for rehabilitating and evaluating this population (Alkjaer et al., 2002; Mattacola et al., 2004). The activation values in this study often exceeded 100% MVIC, an observation that is not uncommon (Jonhagen et al., 1996; Benoit et al., 2003). Because the subjects in this study were motivated trained athletes experienced in performing to maximal exertions and were given encouragement to perform maximal contractions during the isometric tests, we are confident that they exerted themselves to the maximum of their ability during MVIC testing. The high levels of activation during the lunges therefore indicate that the isometric tests may not have been optimal for eliciting a maximal activation or simply that activation levels during such a ballistic movement will elicit higher activations than isometric contractions. Because the comparisons were repeated measures on the same subject using the same normalization technique, the comparative analysis of the EMG signals of the two motions is valid.

The subjects in this study performed the lunges with a maximal effort. When attempting to use these lunge exercises for training or rehabilitation, the difficulty of both lunges, and the JFL in particular, should be gradually increased. Furthermore, given the relatively high levels of activation found during the JFL, it may be more appropriate to begin training with the WFL and progress to the JFL in order to minimize the risk of injury.

Perspectives

We have studied the activation patterns of three of the large muscle groups of the lower leg during the forward lunge. Based on our analysis, we found that the quadriceps contract eccentrically during the first part of a forward lunge, especially the JFL. The hamstrings have an isometric contraction during the initial period of the stance phase, and the gastrocnemius works eccentrically during most of the stance phase. Thus eccentric, concentric and isometric contractions are found during the forward lunge in all three muscle groups. Contrary to popular belief in the track and field community, the forward lunge is not an exercise that elicits prolonged eccentric hamstring activity. More studies need to be conducted to elucidate whether forward lunges can prevent and or treat musculo-tendinous injuries to the quadriceps and hamstrings.

Key words: forward lunge, eccentric, lengthening, quadriceps, hamstring, gastrocnemius, EMG, motion analysis.

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